

Organizing Found Information in Public Digital Library Search

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Abstract

When undertaking complex search scenarios within public digital libraries, users often face challenges in organizing found information. This is an important aspect in supporting exploratory search and cross-session search. In this paper, we present a novel public digital library search interface that includes nested workspaces and a generative AI approach to summarizing what has been saved in these workspaces. This semi-automatic approach to organizing found information was designed to support (1) metacognitive aspects of planning, monitoring, and evaluation during exploratory search and (2) task resumption during cross-session searching. Enhancing workspaces beyond simple collections of saved information holds promise for enabling searchers to more effectively make sense of what they have found while undertaking cross-session exploratory searches.

CCS Concepts

• **Information systems** → **Search interfaces**; **Digital libraries and archives**.

Keywords

exploratory search, cross-session search, public digital library, nested workspaces, summarization, generative AI, metacognition, task resumption

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1 Introduction

In public digital libraries, users may engage in exploratory search activities when faced with complex search scenarios [37]. Such exploratory searching involves multiple stages of gathering, organizing, and synthesizing found information and resources. As such searching takes time and focused attention, it may be difficult to complete such searches in a single session, leading to the need for cross-session search support [21]. However, existing library interfaces often fall short in supporting such search activities, with

limited capabilities for handling exploratory search [10] and for managing collected information [22].

Searchers face significant challenges in organizing found information, tracking their search history, and integrating new knowledge. This is especially true in exploratory search, where searchers are trying to solve complex problems and gathering information from multiple resources. Without structured tools, searchers often rely on manual methods, such as reviewing lists of saved resources or personal notes, which can be inefficient and disruptive to their search process. To address this problem, we pose the following research question:

RQ: How can we design and implement a public digital library search interface that allow searchers to organize and manage found information in support of cross-session exploratory search?

As an answer to this question, we present a Workspace-Enhanced Search interface called WESearch. WESearch makes two primary contributions to the design of digital library search interfaces: a nested workspace framework that allows searchers to dynamically organize saved resources at the task and sub-goal levels, and an automatic summarization mechanism that leverages modern generative AI techniques. WESearch has been designed to support a broad audience of public digital library searchers, preserving the core search interface paradigm to maintain familiarity, and adding lightweight additions that are interactive and persistent, to enhance the search process [17].

The features of WESearch were designed to empower users in the metacognitive aspects of planning, monitoring, and evaluating, which are important considerations in relation to exploratory search [9]. Related to this, they explicitly support the creation and pursuit of sub-goals during the exploratory search process [33]. These have the added benefit of supporting cross-session searching and the fundamental task of reacquaintance [14, 21].

In the remainder of this demonstration paper, we will provide a brief overview of the extant literature, explain the core features of WESearch, outline the technical aspects of the work, and conclude with a discussion of the contributions, limitations, and future work.

2 Literature Review

2.1 Complex Search

An information need occurs when users realize a gap in their knowledge and seek information to bridge it. Traditional information retrieval (IR) systems are effective when needs are well-defined, as they assume accurate queries and matches to the results. However, when users face complex search tasks, such as learning about a topic, their information need may be ill-defined, open-ended, or multi-faceted [37]. In such cases, traditional IR systems are not a

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good match for the complexity of the task [26]. Further, as the pursuit of complex tasks requires the searcher to issue multiple queries and actively engage in the search process, it may not be possible to complete the task in a single session, leading to cross-session searching [21].

2.2 Exploratory Search

Marchionini [23] described exploratory search as a process comprised of three types of search behaviours: lookup, learn, and investigate. Building upon this early work, White and Roth suggested that exploratory search consists of two high-level phases: exploratory browsing and focused searching [37]. Generally, a searcher undertaking complex search tasks will start with the exploratory browsing phase, and then proceed to focused searching when they have learned enough about the topic to reduce their uncertainty and focus on specific aspects of interest [18].

White and Roth's model specified three primary exploratory browsing activities: discovery, learning, and investigation [37]. Discovery is a process of encountering and identifying previously unknown information. Learning is a process of internalizing and making sense of the discovered information, leading to the acquisition of new knowledge and understanding. Investigation is a higher-order cognitive activity that focuses on the analysis, synthesis, and critical assessment of what has been discovered and learned.

Once enough information has been gathered, the searcher may shift to focused searching, which includes three key activities: query (re)formulation, result examination, and information extraction [37]. Query (re)formulation involves iteratively refining the search query to align with the current focus of the search. Result examination requires evaluating the relevance of individual search results. Information extraction refers to selectively identifying and saving useful material from the information source. As the search becomes more focused, the searcher's ability to differentiate between relevant and irrelevant information improves [20].

2.3 Metacognition

The term metacognition, coined by Flavell [12], is often simplified as "thinking about thinking" or "cognition about cognition". Barak [2] suggests that metacognition involves the use of strategies to control cognitive activities in order to meet a particular goal.

There is general agreement that metacognition is comprised of two key aspects: metacognitive knowledge and metacognitive regulation [12, 16, 24, 29]. Metacognitive knowledge refers to knowledge about, and awareness of, one's own cognitive processes. Metacognitive knowledge (or knowledge of cognition) contains three kinds of knowledge: declarative knowledge, procedural knowledge, and conditional knowledge [3, 19, 27, 28]. Metacognitive regulation refers to one's procedural knowledge for regulating cognitive processes, drawing upon an individual's knowledge (about self and strategies, including how and why they use particular strategies) and uses executive skills (such as planning, self-correcting, goal setting) to optimize the use of their own cognitive resources. Regulation of cognition consists of the activities used to regulate and oversee learning [3]. These processes include planning activities (predicting outcomes, scheduling strategies, and various forms of vicarious trial

and error) prior to undertaking a problem; monitoring activities (monitoring, testing, revising, and re-scheduling one's strategies for learning) during learning; and evaluating outcomes (checking the outcome of any strategic actions against criteria of efficiency and effectiveness).

2.4 Self-Regulated Learning and Sub-Goals

In recent work, Urgo & Arguello [34] explored how learning objectives influence searchers' perceptions and behaviors using Anderson and Krathwohl's taxonomy [1]. To uncover how learning occurs during search, understanding the process of self-regulated learning is critical [32]. Self-regulated learning involves learners actively managing their own learning to achieve goals [29, 38]. The Winne & Hadwin model of self-regulated learning consists of four phases: (1) task definition, (2) planning and goal-setting, (3) studying tactics, and (4) adaptation. Building on this work, Greene et al. [15] developed a self-regulated learning coding framework that specifies five macro-processes: (1) planning, (2) monitoring, (3) strategy use, (4) task difficulty and demands, and (5) interest.

Within these macro-processes, micro-processes of self-regulated learning exist in relation to sub-goals [32]. For example, sub-goals are a micro-process of planning, and monitoring one's progress toward a sub-goal is a micro-process of monitoring one's progress toward the overall goal. Thus sub-goals are smaller, intermediate objectives that help in managing and achieving a larger goal.

For example, when users are searching for information to understand a concept like "Egyptian Civilization", a sub-goal might be to "know about people" or to "identify culture". These sub-goals help users organize their search efforts, monitor their progress, and make the learning process more structured [32].

2.5 Task Decomposition

A task is generally considered as a set of connected physical, cognitive, and affective actions through which individuals try to accomplish some goals in their work or everyday lives [5, 35]. Recently, Shah et al. [30] studied how one might align search processes with task conceptualization. They explored how tree-based structuring of search around tasks could improve system support and interaction, while also considering the potential challenges and trade-offs involved in integrating search more deeply with task-oriented frameworks. The concept of tasks as trees [30] emphasizes hierarchical structures in search tasks, breaking down a complex macro task [7] into sub-tasks, sub-sub-tasks, and actions. Drawing from Byström and Järvelin's models [6] and Xie's hierarchies [39], a high-level task like "arrange a vacation" is decomposed into smaller, more actionable tasks (e.g., booking flights or hotels). These tasks may not always be explicit or planned, and search systems can offer varying levels of support for different layers of the hierarchy. Task trees help reduce cognitive load by breaking down tasks into smaller, manageable goals, allowing search systems to offer more tailored assistance based on the user's underlying intentions [30].

3 WESearch

To address the challenges of cross-session exploratory search posted in the research question, we present a workspace-enhanced search

interface: WESearch. In the subsections that follow, the key features of the interface are explained.

To support metacognitive activities, we have identified four key user-centered actions: (1) initiating a new search task with a dedicated workspace for planning the search activity; (2) resuming prior search tasks by accessing saved workspaces; (3) using AI-generated summaries for quick reacquaintance and monitoring of progress across sessions; and (4) organizing the found information into nested workspaces to support evaluation and understanding of complex topics. The concept of sub-goals guides user interactions in this interface, allowing searchers to set and manage specific objectives within their broader goal (i.e., create workspaces and nested workspaces) and track their progress at both levels.

3.1 Workspace Creation

A necessary first step in supporting the organization of information while undertaking an exploratory search is to create a workspace. Figure 1 shows the top-level workspace management tools of WESearch, organized in a simple card-based layout.

Searchers can initiate a new task using the New Workspace card ❶. If they have previously created a workspace to manage a search interest, they can select an existing workspace card ❷. Each workspace card includes search activity statistics as well as an automatically generated summary of what has been found (which will be described later in this paper). When a workspace is selected, it is highlighted with a bounding and identified under the query box at the top of the page. Any queries issued from this point forward are issued in the context of the selected workspace ❸. The eye icon on the workspace card provides a short cut to viewing the collection of saved resources for the given workspace. A searcher can return to this top-level view of the workspaces at any time by clicking the library logo in the top-left corner of the interface.

As an organizational structure, the workspaces support metacognitive planning, helping searchers structure their task within a dedicated space from the start. They also support task resumption, by providing clear distinctions between a searcher's various prior search interests, and enabling the review of what was previously saved.

3.2 Search and Save in a Workspace

After a query is issued, the workspace panel ❹ is presented alongside a typical search engine results page interface ❺ as shown in Figure 2. When evaluating the search results, any that are deemed to be of interest can be saved to the workspace with the bookmark button ❻. These saved resources can be accessed by clicking the "Workspace Resources" button at the top of the search results list.

The workspace panel ❹ provides an overview of key information related to the task. Initially, the information will be sparse. However, as the searcher issues queries and saves resources, it will be populated with statistics about these activities ❼ and a summary of what has been saved ❽ (see Figure 3). Further, the workspace panel includes a section for nested workspaces and a button to create such organizational structures ❾.

The workspace panel supports metacognitive monitoring, enabling a searcher to easily assess their search progress. Further,

during task resumption, it provides valuable information to enable reacquaintance with the overall task.

3.3 Nested Workspaces

A more detailed view of the workspace panel and nested workspaces is provided in Figure 4. The process of creating nested workspaces allows searchers to break down their overall search goal into smaller, more focused sub-goals. For instance, in a workspace on Egyptian Civilization, searchers might create nested workspaces for topics like "People" or "Language" ❿. This enables explicit metacognitive planning and hierarchical task decomposition.

A progress-tracking feature ❶❶, allows searchers to set the status of each nested workspace as "In Progress", "Deferred" or "Complete" as a mechanism of documenting their metacognitive evaluation of the current state of the sub-goal. Colour coding of these statuses helps searchers to visually track their progress for each sub-goal, allowing them to easily identify areas need more attention. This particular feature also supports the reacquaintance aspect of cross-session searching.

3.4 Search and Save in a Nested Workspace

After searchers have created a new nested workspace or selected an existing nested workspace, all subsequent searching is done within that structure (see Figure 5). The workspace and nested workspace are identified under the query bar ❶❷. Any search results saved while the nested workspace is selected are saved within that structure. A summary of what has been saved is provided to help the searcher with metacognitive monitoring of their progress towards the sub-goal represented by the nested workspace ❶❸.

3.5 Resource Management

As users discover resources, the system ensures easy management by mapping all queries and saved resources to the currently selected workspace or nested workspace. At any point during the exploratory search process the searcher may view what has previously been saved within these organizational structures (see Figure 6).

If a searcher identifies that a resource is also aligned with a different nested workspace, it can be added there too with the contextual menu ❶❹. Doing so adds the nested workspace as a tag to the saved resource ❶❺. Clicking the x beside any tag will remove the resource from that nested workspace.

This process also allows a searcher to create a new nested workspace while evaluating a previously saved resource. Doing so occurs as part of metacognitive evaluation, transitioning to metacognitive planning and the creation of a new sub-goal. The flexibility of the nested workspace structure allows and encourages such restructuring, which is a fundamental part of learning as one searches.

3.6 Summaries

To aid searchers in monitoring their progress, the system automatically generates summaries at both the nested workspace level and the workspace level. At the nested workspace level, these summaries serve as a mechanism for providing an overview of what has been saved at that level of the organizational structure, offering insight into the state of their associated sub-goals. At the workspace level, these summaries provide an overview of the nested

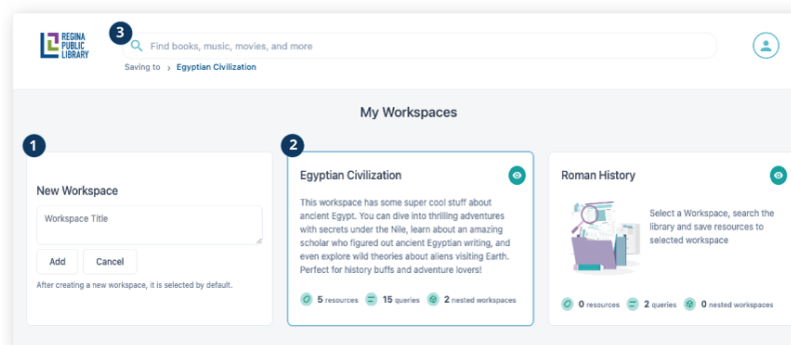


Figure 1: The home page allows a searcher to (1) create new workspace or (2) select an existing workspace. Once a workspace is selected, the searcher may (3) issue a new query to start searching within that workspace.

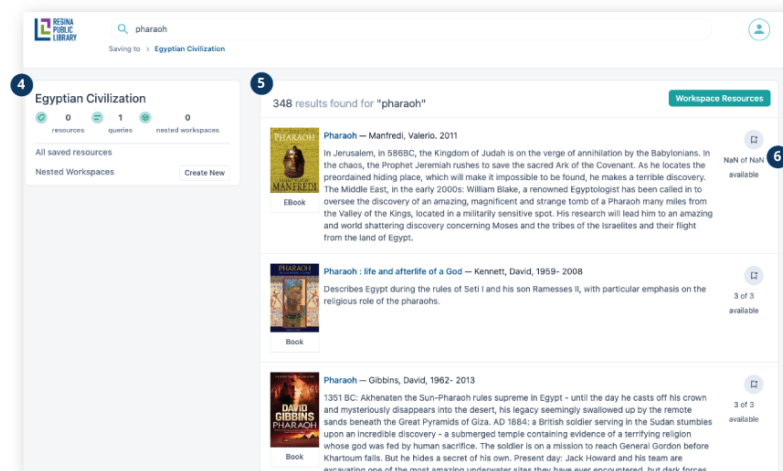


Figure 2: After issuing a query, (4) an overview of the workspace is provided along with (5) a list of search results. The searcher can easily save a resource to the workspace by (6) clicking on the bookmark button for a given resource.

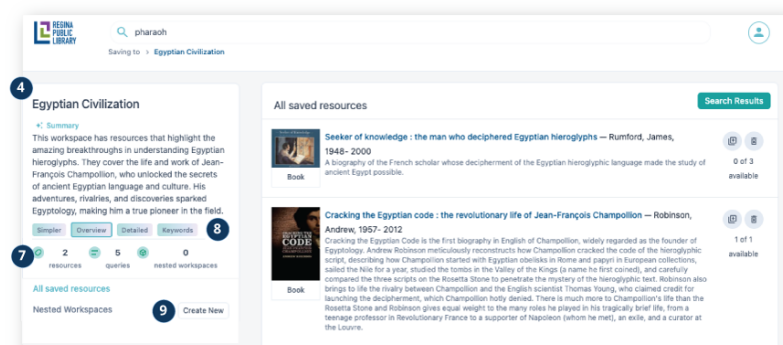


Figure 3: As search results are added to the workspace, the (4) workspace panel, the (7) statistics are updated along with the (8) summary. Nested workspaces can also be created here with the (9) Create New button.

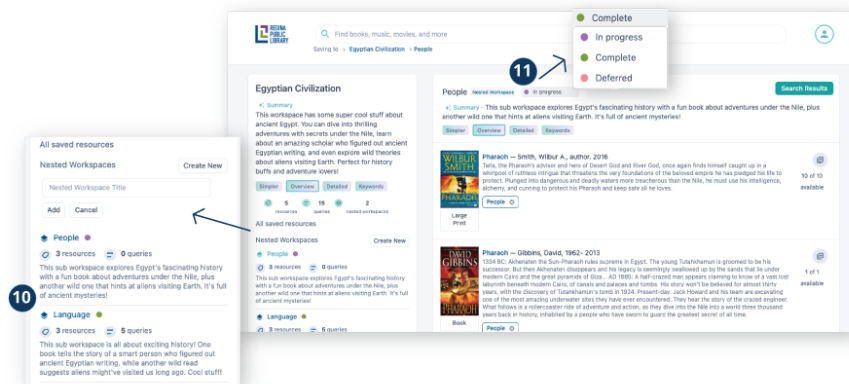


Figure 4: The workspace panel includes information about the (10) nested workspaces as well as a mechanism for (11) progress-tracking for each nested workspace.

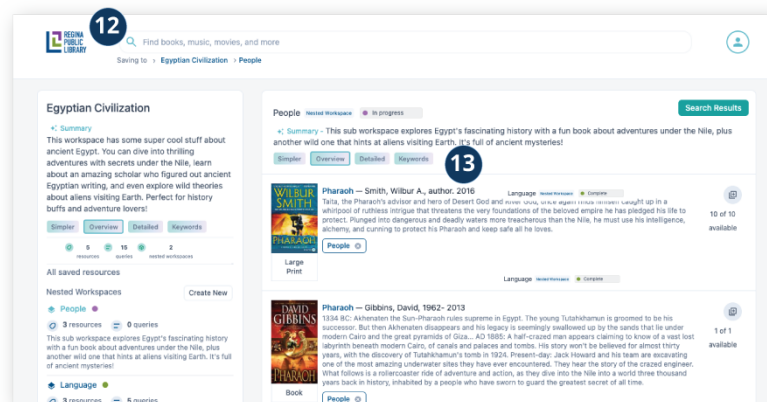


Figure 5: When a nested workspace is selected, (12) all searching done is saved at that level of the organizational structure. (13) Summaries of the nested workspace are also provided.



Figure 6: Viewing the saved resources in the workspace allows searchers to (14) add resources to other nested workspaces and to (15) view the complete set of nested workspaces for a given resource.

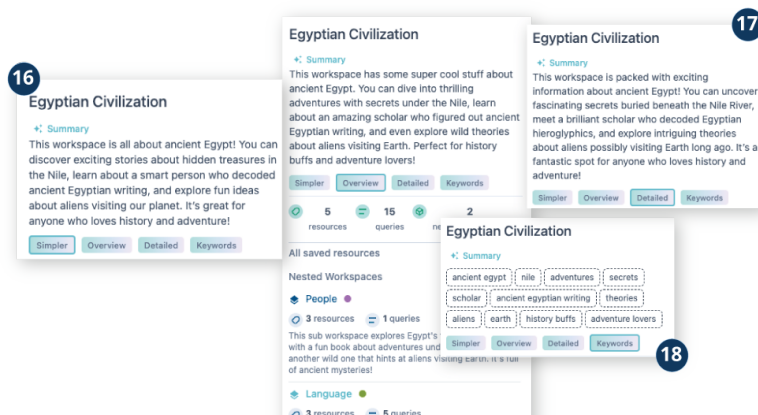


Figure 7: The default overview can be changed to be (16) simpler, (17) detailed, or (18) composed only of keywords.

workspaces, which is aligned with the overall search goal. Whenever a search result is added to a workspace or nested workspace, the summaries are re-generated to reflect the new information saved within the organizational structure.

By default, the summaries are generated to provide an overview of the respective level of the organizational structure. As illustrated in Figure 7, the searcher has the option to request a simpler summary 16, a more detailed summary 17, or to represent the summary as keywords 18. The completeness or incompleteness of these summaries with respect to the goal (workspace) or sub-goal (nested workspace) provide cues to support metacognitive monitoring.

4 Implementation

4.1 Software Architecture

WESearch is implemented as a web-based application built with ReactJS and styled using Tailwind. Requests from the front-end are handled by a Node.js back-end. The public library search results are provided by an API from the Saskatchewan Information & Library Services Consortium. User data and saved resources are stored in a MongoDB database.

4.2 Internal Architecture

The application features a two-level hierarchical workspace structure, where a top-level workspace may contain multiple nested workspaces, each storing saved resources. Summaries are generated at all levels, but only the workspace-level summaries are displayed to users to prevent information overload.

4.3 Generative AI and Summaries

A generative AI approach is used to create the summaries. This is done at the saved resource level, the nested workspace level, and the workspace level (see Figure 8). Whenever a search result is saved, its metadata is passed to the OpenAI API with a prompt requesting the information to be summarized. Nested workspace summaries are created in a similar way, based on the summaries of the saved resources. In order to provide an overall view of a

top-level workspace, its summary is based on the summaries of all saved resources and the summaries of the nested workspaces.

A structured approach to prompting is crucial for enhancing the quality and reliability of large language model (LLM) and ChatGPT outputs [31]. By clearly defining components such as Task, Instructions, and Role, users can effectively guide the AI, aligning its responses with specific objectives to ensure outputs are relevant, clear, and actionable. Other recommendations for prompt design have noted that key elements for clarity include specific task instructions, appropriate context, and strategic wording [11]. A “building block” approach [11] to structuring prompts may specify details such as “task type”, “tone”, and “style”. Our work on prompt design has also been influenced by guidance from OpenAI [25] and the Nielsen Norman Group [4].

Following this guidance, we start by specifying constraints to ensure concise, contextually rich, and cohesive summaries. That is, we specify word limits on the summaries (20 words for the resources, 30 words for the nested workspaces, and 50 words for the workspaces). We also specify a limit on key concepts at the nested workspace level and features to extract, as well as a limit on the common themes at the workspace summary level.

As summarized in Figure 9, the specific prompt design specifies a task, format, persona, context, and tone. It also instructs the AI to generate the summary based on the resources specified. The **task** specifies that the generative AI should produce a summary within a specified word limit, promoting focus and brevity. The **format** requires a paragraph structure, enhancing readability. The **persona** specifies an 8th grade reading level, as representatives of general public users with basic public digital library skills to ensure the language is appropriate for their comprehension level. The **context** provides a clear purpose by indicating that the summary is intended for gathering information for an assignment, further reinforcing the need to summarize. Lastly, the **tone** asks for clear and friendly language, making the summary accessible and engaging for all users.

When the searcher chooses the simpler, detailed, and keyword options for the summaries, the prompts are modified with respect to the persona (decreasing or increasing the reading level) and format (keyword list instead of paragraph).

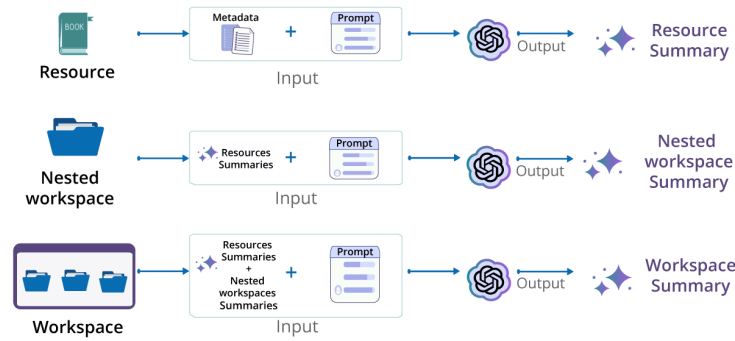


Figure 8: The summaries are created at the individual resource, nested workspace, and workspace level, using a generative AI approach.

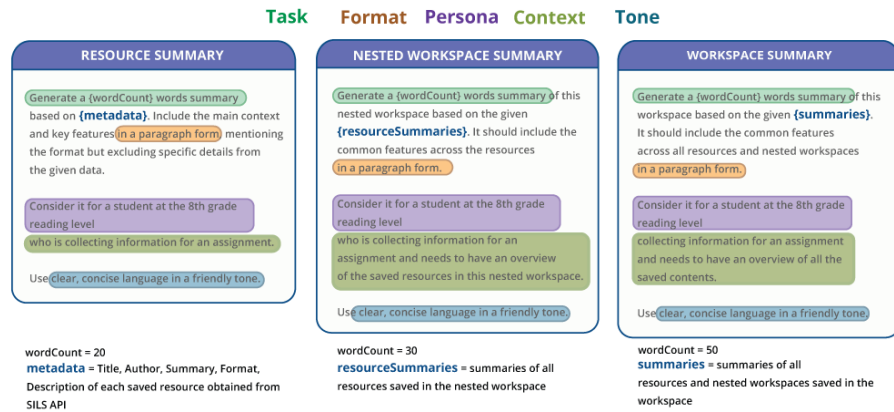


Figure 9: Structure of prompt design at the resource, nested workspace, and workspace levels.

5 Scenario of Use

A supporting scenario of use video has been submitted to the ACM Digital Library, illustrating how WESearch aids users in conducting exploratory searches across multiple sessions. It shows how a user can plan and create workspaces and nested workspaces to organize their search, save relevant resources, and use the summaries to monitor their progress. The video highlights the seamless experience of returning to a previously searched topic, allowing the user to evaluate their findings and adjust their search strategies to meet their search goals and sub-goals.

6 Discussion

WESearch provides a semi-automated approach to organizing found information within a cross-session exploratory search context. Searchers are able to use their human intelligence to plan their search with the workspaces and nested workspaces, and to save relevant resources within this organizational structure. It uses generative AI to summarize what has been found, supporting the searcher to monitor their progress.

This work is fundamentally different from other approaches for leveraging what has been saved to a workspace in the process of sensemaking. While most provide workspaces as simple

lists of saved resources, some recent work has attempted to make such workspaces interactive with the goal of enabling sensemaking. OrgBox provides an interactive note-taking approach in which searchers can create a two-level hierarchical structure of notes through indenting. [8, 9]. KLink Search provides an interactive keyword highlighting approach to allow searchers to surface relationships among saved resources through common keyword use [18]. Like WESearch, both approaches require that searchers first save relevant resources to the workspace. Where WESearch differs is that the support for sensemaking provided by the summaries is entirely automated.

This work has a relationship to retrieval-augmented generation (RAG) [13, 36]. RAG approaches generate answers based on a set of search results retrieved from a collection. Doing so relies on the user providing a precise description of what they are seeking and the information retrieval engine ranking the results accurately. Our approach does not rely on such constraints, basing the summary only on the resources the searcher has explicitly chosen as relevant using their human intelligence. Since the summaries generated in WESearch are based on user-generated workspaces, we classify this work as workspace-augmented generation (WAG).

While WESearch was designed to enhance cross-session exploratory search, it is not without its limitations. The process of creating workspaces and nested workspaces as part of the planning and sub-goal setting takes time and effort. We mitigate this by making the process as simple as possible and by allowing the nested workspaces to be created in the midst of the search process rather than requiring that they be specified in advance. As the summaries are generated based on what is saved to the nested workspaces, the quality is dependent on the quality of the saved resources and the quality of their metadata. Finally, the non-deterministic nature of generative AI approaches means that the summaries generated may be different each time. Precise prompt design and consistent saved resource metadata minimizes the impact of this limitation.

We are in the process of conducting a user study that compares WESearch to a baseline system that includes note-taking features to support the organization of the found information. We are also investigating methods for expanding the generative AI approach to provide information extraction, which is a critical final activity in the exploratory search model. Finally, we are considering how this approach might support cross-session exploratory search in the context of academic search.

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